1796–2020 • vaccine innovation, elimination, herd immunity, remaining gaps

The Power of Vaccination:

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines transformed public health by eliminating or drastically reducing deadly diseases over two centuries.

1796

Edward Jenner created the first vaccine against smallpox.

<1 year

COVID-19 moved from pathogen identification to vaccine licensure in 2020.

1980

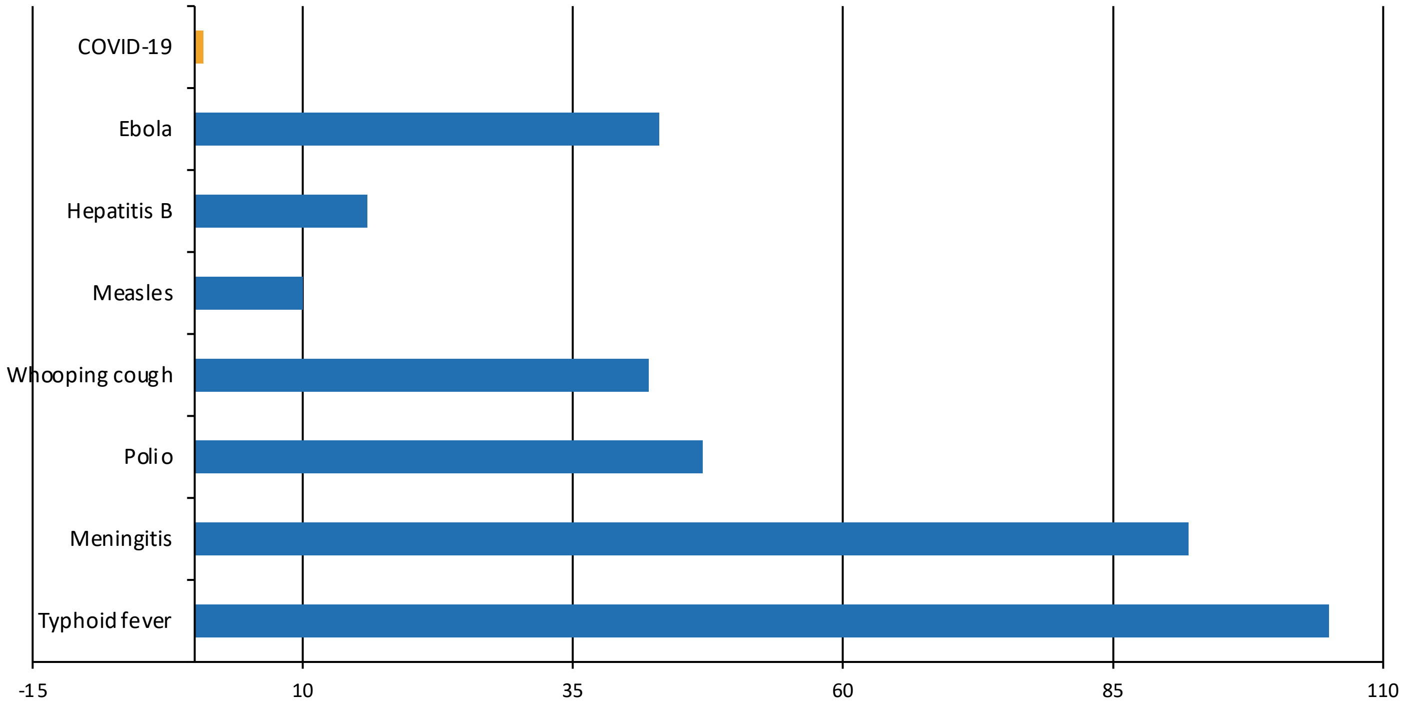
Smallpox declared eradicated worldwide.

Audience: public health professionals and policy analysts

Source: Our World in Data vaccination brief

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

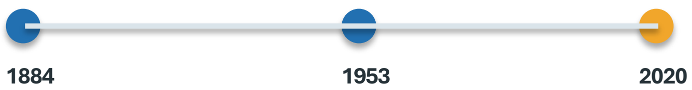
Gap from pathogen identification to vaccine licensure (years)



COVID-19 is visually emphasized: pathogen identified and vaccine licensed in 2020, a gap of under 1 year.

Modern tools cited in the brief — advanced microscopy, genome sequencing, mRNA technology, and virus-like particles — enabled faster and more precise vaccine design.

Pathogen identified → vaccine licensed



Early vaccine gaps were measured in decades; COVID-19 compressed the interval to record time.

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-vaccine cases /M/yr	Post-vaccine cases /M/yr	Case reduction	Death reduction
Diphtheria	**158**	**0**	**100%**	**100%**
Smallpox	**250**	**0**	**100%**	**100%**
Acute polio	**141**	**0**	**100%**	**100%**
Measles	3,044	0.2	**99.99%**	**100%**
Rubella	242	0.04	**99.98%**	**100%**
Pertussis	1,534	52	**96.6%**	**99.7%**
Hepatitis A	465	51	**89%**	**88.7%**
Acute hepatitis B	273	44	**83.9%**	**83.6%**
Pneumococcal	233	139	**40.5%**	**31.3%**

Diseases marked in green reached 100% reduction in both cases and deaths in the US data table: diphtheria, smallpox, and acute polio.

Remaining burden is concentrated in pneumococcal disease, hepatitis A, and acute hepatitis B — the same gap areas revisited later.

100%

US case and death reduction for diphtheria, smallpox, and acute polio.

40.5%

Lowest case reduction shown: pneumococcal disease.

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Before 1963, nearly every state experienced high measles incidence: 100–1,000+ cases per 100,000.

After the 1980 kindergarten vaccination mandate, cases dropped to near zero across all states, with only sporadic outbreaks.

Annotated milestones

- 1963: First measles vaccine developed by John Enders
- 1971: MMR vaccine developed by Maurice Hilleman
- 1980: Vaccination mandatory for kindergarten entry

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

Smallpox: The Only Human Disease Eradicated Through Vaccination

1959
WHO resolution to eradicate smallpox

1967
Intensified global campaign launched

1977
Last natural case in Somalia

Declared eradicated worldwide

Smallpox was declared eradicated worldwide in 1980 after a global campaign — the greatest vaccination success story in the brief.

1980
certified eradication

Only one

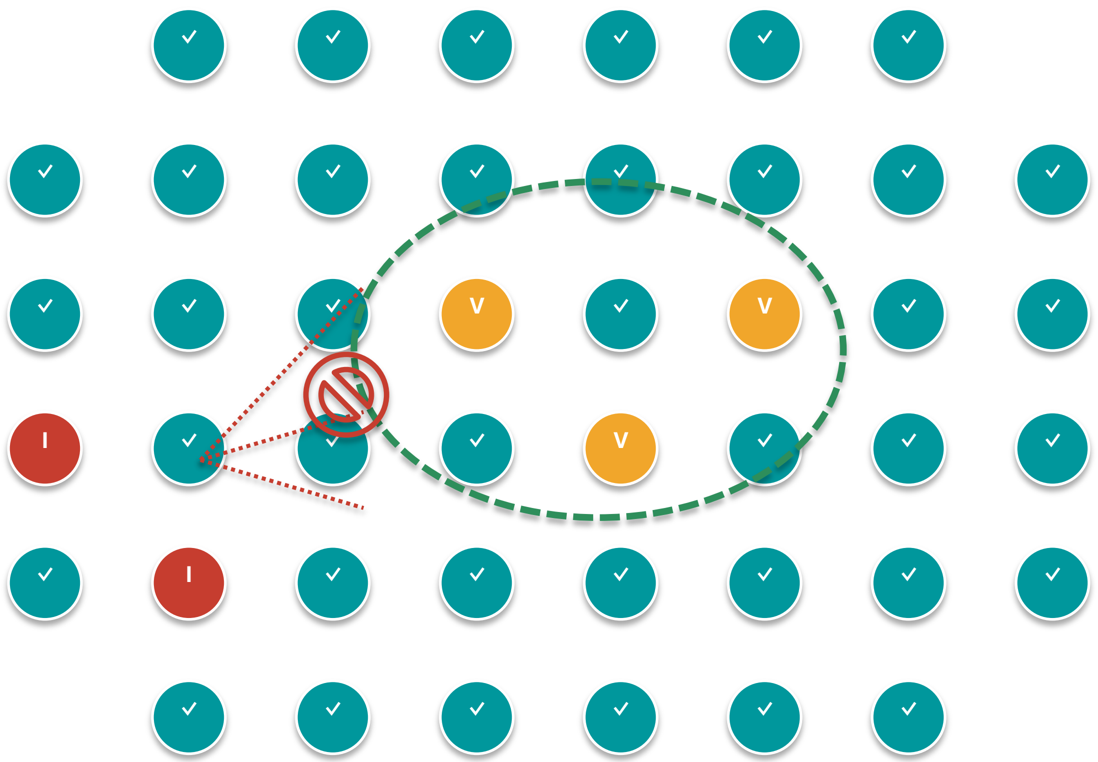
Smallpox remains the only human disease fully eliminated through vaccination.

Why eradication was feasible

- The virus infected only humans
- Symptoms were distinguishable
- The vaccine was cheap and effective

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Vaccinated people reduce transmission pathways

When enough people are vaccinated, pathogens cannot spread effectively.

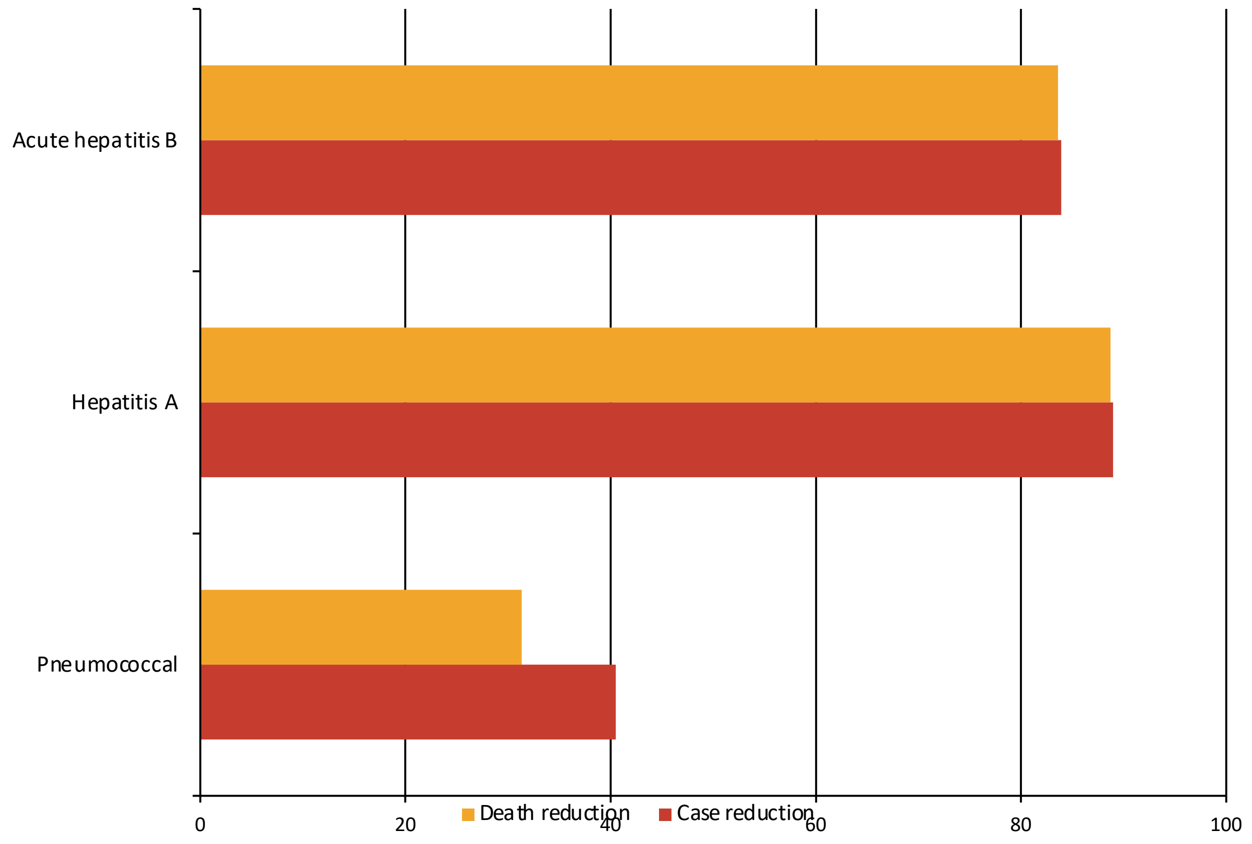
Community protection shields those who cannot be vaccinated: newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Policy implication

- Coverage is a population-level intervention
- Local gaps can reopen transmission chains
- Trust, access, and supply are core health-system functions

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



40.5%

Pneumococcal case reduction — the lowest case-reduction figure in the brief.

31.3%

Pneumococcal death reduction.

Barriers named in the brief

- Not everyone globally has access to vaccines
- Supply shortages put children at risk
- Poor scientific understanding can reduce trust and uptake

The next frontier is not only invention — it is reliable delivery, access, supply resilience, and public trust.

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job ~~Isn't Done~~

1. **Elimination is possible**

   Diphtheria, smallpox, and acute polio reached 100% reduction in US cases and deaths.

2. **Development has accelerated**

   Timelines collapsed from typhoid fever at 105 years to COVID-19 at under 1 year.

3. **State-level impact is visible**

   Measles moved from 100–1,000+ cases per 100,000 before 1963 to near zero after the 1980 mandate.

4. **Eradication has precedent**

   Smallpox was declared eradicated worldwide in 1980 after the WHO resolution, intensified campaign, and last natural case.

5. **Coverage gaps remain**

   Pneumococcal, hepatitis A, and acute hepatitis B reductions remain incomplete.

Call to action for global health systems

Improve access, coverage, supply resilience, and trust in vaccination — so innovation translates into population protection.

Innovation + delivery + trust

End state: stronger routine immunization and faster, more equitable responses to emerging pathogens.

Source: Our World in Data; Roush & Murphy (2007) data as cited in source brief